

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter 2024

Course: B.Tech

Branch : Chemical Engineering

Semester :III

Subject Code & Name: BTCHC304- Mechanical Operations

Max Marks: 60

Date:12/ 02/ 2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	For coarse reduction of hard solids, use	2/CO1	1
	a Impact b Attrition c Compression d Cutting		
2	The maximum slope of a belt conveyor can be	3/CO1	1
	a 15° b 30° c 45° d 60°		
3	Screen efficiency is----	2/CO2	1
	a)recovery/rejection b)recovery c)rejection d)none of these		
4	In the cyclone separator used for separation of dust from dust laden gas, the gas	3/CO3	1
	a) enters the cyclone from the top b) is admitted tangentially at high velocity c) develops a helical motion inside the chamber d) Both (b) and (c).		
5	Which of the following is not used as filter aid?	2/CO4	1
	a)Asbestos b)Diatomaceous earth c)Purified wood cellulose d)Rice husk		
6	During filtration operation, the filtrate encounters the resistance of the	3/CO3	1
	a)Filter medium b)Cake c)Channel carrying the slurry d)All (a), (b) and (c)		
7	Which of the following is not an application of transport in membranes?	3/CO4	1
	a)Microfiltration b)Reverse osmosis c)Dialysis d)Fractional distillation		
8	Which is the most practical and economical method for removal of Suspended solid matter from polluted water?	3/CO5	1

	a) Sedimentation	b) Skimming off	c) Chlorination	d) Biological oxidation		
9	_____ is the most suitable for compounding rubber and plastic solids.				3/CO1	1
	a) Banbury mixer	b) Pan mixer	c) Pug mill	d) Charge can mixer		
10	Jigging is a technique by which different particles can be				2/CO1	1
	a) Separated by particle size	b) Separated by particle density	c) Separated by particle shape	d) Mixed		
11	Which of the following is not used as filter aid?				2/CO3	1
	a) Asbestos	b) Diatomaceous earth	c) Purified wood cellulose	d) Rice husk		
12	A fluid energy mill is used for				2/CO1	1
	a) Cutting	b) Grinding	c) Ultra grinding	d) Crushing		
Q.2	Solve the following.					12
A)	Calculate the operating speed of the ball mill from the following data 1) Diameter of ball mill=800mm. 2) Diameter of balls = 60mm If Operating speed is 55% less than critical speed				4/CO1	6
B)	Explain the working of Ribbon blender with well labeled diagram.				3/CO1	6
Q.3	Solve the following.					12
A)	Explain the working of Vibrating Screen with well labeled diagram.				3/CO2	6
B)	Which are the factors which reduce capacity and effectiveness screen.				3/CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Describe with neat sketch working of rotary drum filter.				3/CO3	6
B)	Explain the operating cycle of centrifuge.				3/CO3	6
C)	A plate and frame press, filtering a slurry gave a total of 8 m ³ of filtrate in 1800 sec and 11m ³ in 3066 sec, when filtration stopped. Estimate the washing time if 3 m ³ of wash water used. The resistance of the cloth can be neglected and a constant pressure is used throughout.				4/CO3	6
Q.5	Solve Any Two of the following.					12
A)	Explain Membrane processes based on size of component in the feed.				3/CO4	6
B)	Explain Ultrafiltration with its application, process driving force and membrane material used. https://www.batuonline.com				3/CO4	6
C)	Define Membrane separation processes with its industrial application over advantages over other unit operations.				3/CO4	6
Q. 6	Solve Any Two of the following.					12
A)	Explain the process of batch sedimentation with diagram.				3/CO5	6
B)	With a neat sketch, describe the working of a DORR THICKENER.				3/CO5	6
C)	With suitable examples, explain the role of coagulants in filtration.				3/CO5	6
	*** End ***					